

AGES

1. The ages of A and B are in ratio 6:5.

After 9 years the ratio of their ages will be 9:8. Find the difference between their ages at present.

ANS: 3 years

$$\frac{6x+9}{5x+9} = \frac{9}{8}$$

$$6x - 5x = 18 - 15 = 3 \text{ years}$$

$$x = 3$$

$$6:5$$

$$\underline{9:8}$$

2. 4 years ago the ratio of ages of Seetha and Geetha was 5:6 and the sum of their ages at present is 52 years. What is the ratio of their present ages?

ANS: 6:7

$$\begin{array}{l} S \rightarrow 5x+4 \\ G \rightarrow 6x+4 \end{array} \left. \vphantom{\begin{array}{l} S \rightarrow 5x+4 \\ G \rightarrow 6x+4 \end{array}} \right\} \text{Present}$$

$$= 5 \times 4 + 4 = 24$$

$$= 6 \times 4 + 4 = 28$$

$$24:28$$

$$\boxed{6:7}$$

$$5x+4 + 6x+4 = 52$$

$$11x+8 = 52$$

$$x = 4$$

3. The present ages of Raju and his Father is in the ratio 2:5. Four years, hence the ratio will become 5:11. What was the Father's age five years ago.

ANS: 35

$$2:5$$

$$2x:5x$$

$$\frac{2x+4}{5x+4} = \frac{5}{11}$$

$$22x+44 = 25x+20$$

$$x=8$$

$$\Rightarrow 16:40 \xrightarrow{\text{Five years ago}} 16-5:40-5$$

$$11:35$$

$\Rightarrow$ Five years ago

4. The present age of Rohit is 8 times the age of his son. 8 years hence, the ratio of ages of Rohit and his son will become 10:3. Find Rohit's present age.

ANS: 32

$$8x:x$$

$$8(4):(4)$$

$$32:4$$

$$8x+8 : x+8$$

$$\frac{8x+8}{x+8} = \frac{10}{3}$$

$$24x+24 = 10x+80$$

$$x=4$$

5. Vignesh married 8 years ago. Today his age is $\frac{9}{7}$ times the age at the time of his marriage. At present his daughter's age is $\frac{1}{6}$ th of his age. What is his daughter's age after 5 years.

ANS: 11 yrs

$$V = \frac{9}{7}(V-8)$$

$$7V = 9V - 72$$

$$2V = 72$$

$$V = 36 \text{ yrs}$$

$$D = \frac{1}{6} \times 36 = 6 \text{ yrs}$$

$$D+5 = 11 \text{ yrs}$$

The ratio b/w ages of A & B is 4:5 and that of A & C is 5:6. If the sum of ages of all the 3 is 69 yrs. Find the present age of B.

ANS: 25

$$a : b = 4 : 5$$

$$a : c = 5 : 6$$

$$20 : 25$$

$$20 : 24$$

$$A+B+C=69$$

$$20x + 25x + 24x = 69$$

$$x=1$$

$$B = 25(1)$$

$$B = 25$$

	a	b	c
	4	5	6
x	20	25	24
	20	25	24

A's age is $\frac{1}{3}$ rd of B's age, B's age is $\frac{4}{5}$ th of C's age, & C's age is $\frac{3}{5}$ th of D's age. Find A's age if D is 50 yrs.

ANS: 8 yrs

$$D = 50$$

$$C = \frac{3}{5} \times 50 = 30$$

$$B = \frac{4}{5} \times 30 = 24$$

$$A = \frac{1}{3} \times 24 = 8$$

In 10 yrs, A will be twice as old as B was 10 yrs ago. If A is now 9 yrs older than B, find the present age of B.

ANS: 39

$$A = B + 9$$

$$A + 10 = 2(B - 10)$$

$$B + 9 + 10 = 2(B - 10)$$

$$19 + B = 2B - 20$$

$$39 = B$$

9. The sum of ages of 5 children born at the intervals of 3 yrs each is 50 years. Find the age of oldest child.

Ans: 16 yrs

$$\underbrace{x}_{\text{youngest}} + (x+3) + (x+6) + (x+9) + \underbrace{(x+12)}_{\text{oldest}} = 50$$

$$5x + 30 = 50$$

$$x = 20$$

$$x = 4$$

$$x + 12 = 16$$

10. The sum of ages of A, B, C is 93 yrs. 10 yrs ago, the ratio of their ages is 2:3:4. Find present age of C.

Ans: 38 yrs

present-

$$2x + 10 + 3x + 10 + 4x + 10 = 93$$

$$9x + 30 = 93$$

$$9x = 93 - 30$$

$$x = 7$$

$$C \Rightarrow 4x + 10 = 4(7) + 10 = 38$$

11. The ratio of ages of man and his wife is 4:3. After 4 yrs the ratio will become 9:7. If, at the time of their marriage, the ratio of their ages is 5:3 then how many years ago, they got married.

Ans: 12 yrs

$$\frac{4x + 4}{3x + 4} = \frac{9}{7}$$

$$x = 8$$

$$32:24$$

$$\frac{32 - x}{24 - x} = \frac{5}{3}$$

$$96 - 3x = 120 - 5x$$

$$x = 12$$

12. Tanya's grand father was 8 times older to her 16 yrs ago. He would be three times of her age 8 years from now. What was the ratio of ages of Tanya's to her grand father.

ANS: